

Arnav Amit
Independent Researcher
4241 W Kling St, Burbank, CA 91505
arnavamit1@g.ucla.edu | (469) 408-3425

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Editors, Journal of Electrocardiology

Dear Editors,

I am submitting an original research article, "Rate Correction Misrepresents Cardiac Risk in Pediatric Psychiatric Polypharmacy: A Mechanistic Action Potential Analysis Separating Genuine Repolarization Change from Heart Rate Artifact," for consideration in the Journal of Electrocardiology.

Psychiatric polypharmacy is now routine in children and adolescents, yet cardiac risk in this population is assessed almost entirely through Bazett corrected QTc, a metric extrapolated from adults. Using a mechanistic O'Hara-Rudy ventricular action potential model coupled to a three pathway drug module, I show that this metric misrepresents repolarization risk in opposite directions across the two most common rate changing drug classes. It manufactures apparent prolongation for stimulants, whose action potential in fact shortens, and it masks genuine prolongation for alpha-2 agonists, which slow heart rate. Across all 84 combinations, Bazett flagged 29 at or above 10 ms while only 5 carried genuine repolarization change, and the two measures were negatively correlated. Only the uncorrected action potential duration recovers the true direction and magnitude of effect.

I also report a pediatric FAERS analysis of 552,832 cases, including a 23 to 50 fold higher stimulant cardiac reporting signal in children than in adolescents. I am explicit throughout that the model is mechanistic and hypothesis generating, and that its concordance with pharmacovigilance is weak; I make no claim that it is a validated clinical predictor.

I believe the work fits the Journal of Electrocardiology because its central contribution concerns the QT interval and its rate correction, the methodological core of the journal's scope, rather than a drug safety claim alone. The analysis builds directly on normal ECG values work published in this journal.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. It is single authored; I conducted the work as an independent researcher. All code, data, and results, including the validated cardiac engine, the drug effect module, the polypharmacy sweep, and the FAERS pipeline, are openly available at github.com/arnava25/cardiosafe-pediatric. The work received no funding and I have no competing interests to declare. I used a generative AI tool for manuscript drafting and revision, as disclosed in the manuscript, and I take full responsibility for the content.

Thank you for considering the manuscript.

Sincerely,

Arnav Amit